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An Integrated Sensor Network and Data Driven Approach to Satellite Remote Sensing of Dissolved Organic Matter

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Key Points:

- Coupling remote sensing and in situ sensors in an integrated sensor network provides the framework for robust water quality models
- Optical sensors measuring dissolved organic matter fluorescence are good indicators of total dissolved organic matter in a complex estuary
- High spatial resolution satellite imagery is key to developing water quality retrievals for inland and coastal waters

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Traditional remote sensing retrieval models for water quality have historically relied on limited, localized data sets due to the prohibitive costs of extensive field campaigns and logistical challenges of collecting match-up data with satellite overpasses. As a result, these models often lack generalizability across seasons, tides, and sites. Furthermore, small field data sets limit the utility of modern machine learning techniques to advance remote sensing retrieval models. In situ optical sensors deployed in a sensor network to continuously monitor larger water bodies can drastically increase the number of measurements, providing the opportunity to develop new approaches for building robust remote sensing retrieval models by leveraging both remote sensing data and in situ networks as an integrated monitoring system. This study leverages a large "ground-to-space" sensor network that combines an in situ optical sensor network with satellite-based remote sensing to overcome these limitations. Utilizing a large-scale data set from the U.S. Geological Survey's Sacramento—San Joaquin River Delta monitoring network, of dissolved organic matter fluorescence measurements, and remote sensing data from the European Space Agency's Sentinel-2A and -2B satellites, this study implemented a data driven approach for dissolved organic matter models. The data set, consisting of 982 samples collected between 2018 and 2021 was used to train and validate a random forest model ($R^2 = 0.76$, RMSE = 6.1 Quinine Sulfate Equivalents), with demonstrated applicability across diverse site conditions, tidal stages, and seasons. This work provides a scalable solution to address critical challenges in water quality monitoring and offers a replicable framework for global water quality management.

1. Introduction

Optical satellite remote sensing has been established as a popular technology to monitor water quality due to its potential for broad geographic coverage and systematic, repeat sampling (Giardino et al., 2019; Hestir et al., 2015; Mouw et al., 2015; Olmanson et al., 2011; Topp et al., 2020). The emergence of new space-borne multispectral missions with high spatial resolution and repeat frequency, such as the European Space Agency's Sentinel-2 satellites, have resulted in an explosion of applications for optical remote sensing of water quality in coastal and inland water bodies, such as measuring parameters of interest for ecosystem functioning, nutrient cycling, and environmental and public health. These parameters include colored dissolved organic matter (DOM; Brezonik et al., 2015; J. Chen et al., 2019; Olmanson et al., 2016; Pahlevan et al., 2022; Toming et al., 2016), turbidity (Dogliotti et al., 2015; C. M. Lee et al., 2021; Pahlevan et al., 2022), and chlorophyll *a* (Kuhn et al., 2019; Lopez Barreto et al., 2024; Pahlevan et al., 2022).

In this work we demonstrate the utility of an integrated sensor network that builds effective models for DOM retrievals using coupled, extensive in situ data sets and satellite data in the Sacramento—San Joaquin River Delta, an extensive inland river delta and estuary system in California. We used an emerging design concept for robust water quality monitoring: an integrated complementary system of in situ optical sensor networks, which provided high temporal frequency measurements distributed at point locations at relatively low-cost (as compared to absorption-attenuation sensors), coupled with remote sensing imagery (Arias-Rodriguez et al., 2021; El Serafy et al., 2021).

We focused on a data-driven approach to developing a water quality parameter retrieval model, in contrast to a physics-based model. There are pros and cons in each approach. Historically, remote sensing of water quality parameters has relied on a semi-analytical physics-based approach to model development (Topp et al., 2020). However, this approach can lead to challenges in water quality monitoring from satellite-based platforms. Current satellite radiometers that provide the spatial resolution necessary for monitoring inland and coastal water bodies

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are designed for land remote sensing; they are not as well suited for the optical complexity of inland and coastal waters (Giardino et al., 2019; Mouw et al., 2015). Additionally, semi-empirical physics-based approaches still require parameterization that most often comes from in situ data (Malthus et al., 2012).

In contrast, data-driven modeling approaches, while also relying on in situ data, can be developed for a broader range of measurement parameters, including those made via fluorescence, which are relatively low-cost sensors compared to absorption measurements. Thus, a broad optical sensor network of fluorometers provides the advantage of creating a much larger data set of water quality parameters that is more representative of the variability within a system. By leveraging cloud-based computing resources and automated modeling frameworks, this approach then leads to models that are easy to update and incorporate near real-time satellite imagery and in situ measurements, leading to optimized water quality retrieval models for a given system. Additionally, this provides the framework to quickly adapt to, and assess, changing ecosystems in a warming climate.

We focused on DOM due to the growing recognition of the importance of DOM in environmental health and aquatic ecosystems, which has led to increasing interest in measurement technologies (Pellerin et al., 2016) for this parameter. DOM plays a large role in aquatic ecosystem functioning (Tanentzap et al., 2019) and in the carbon transfer from terrestrial to aquatic systems (Casas-Ruiz et al., 2023; Lehman et al., 2015). DOM is a key driver in aquatic ecosystem metabolism (D'Andrilli et al., 2022), controls nutrient availability (Childers et al., 2006; Fork et al., 2020), and controls heavy metal mobility and bioavailability (Bergamaschi et al., 2011, 2012; Tang et al., 2020). The measurement of DOM is usually conducted with optical techniques that rely on the assumption that the optically active portion of the DOM pool is a reasonable proxy of the total DOM content of most water bodies (Hestir et al., 2015). This optically active, or "chromophoric" or "colored" portion of the DOM pool, is often referred to as "CDOM."

Two common in situ optical techniques for measuring CDOM are via spectral absorption (aCDOM) or fluorescence (fDOM) instruments deployed in the water column or using flow-through systems. CDOM absorption and attenuation spectra are then characterized by the amplitude and slope, decreasing exponentially from the ultraviolet (UV) through the visible range of the electromagnetic spectrum (Roesler & Perry, 1995). The magnitude of the absorption coefficient at a given wavelength is used as an indicator of the amount of CDOM in the water (Bricaud et al., 1981) and the slope is considered an indicator of the quality or source of the CDOM (Chen et al., 2011). Instruments have also been developed that measure the fluorescent portion of DOM (fDOM) in blue to green wavelengths when exposed to UV light (Booth et al., 2023). The intensity of the fDOM measurement is related to the amount of DOM in the water (Ruhala & Zarnetske, 2016). The relationship between in situ measurements of aCDOM and fDOM has been explored and a high correlation between fDOM and aCDOM has been established across a range of CDOM concentration values (Nieke et al., 1997; Niu et al., 2014).

In situ fDOM sensors are increasingly popular relative to labor intensive water sample lab analysis or high-cost in situ absorption sensors due to their ease of use and approachable cost (Coble & Gagosian, 1991; Pellerin et al., 2016). The equipment required to measure in situ CDOM absorption is costly and not widely available, and grab sampling for laboratory analyses of absorption is laborious and costly, resulting in small data sets ($n < 100$) over limited regional areas (Chen et al., 2019). This leads to remote sensing-derived CDOM products that are not fully validated for a range of CDOM conditions, because of the small in situ data sets and limited data collected concurrently with satellite overpasses (hereafter referred to as match-up data). These factors contribute to fDOM sensors becoming widely accepted as surrogate for CDOM in the water monitoring community (Downing et al., 2012; Nieke et al., 1997; Ruhala & Zarnetske, 2016) because of their perceived advantage over labor intensive water sample laboratory analysis or higher cost absorption sensors. While in situ DOM monitoring can provide high fidelity measurements at rapid repeat intervals that capture the variability of aquatic environments (often near-continuous monitoring at 5- or 15-min increments), they are often limited in the geographic scope. Even a network of lower cost sensors leaves gaps in spatial coverage that limit the ability to make inferences about the transport and fate of DOM in aquatic ecosystems. The integrated sensor network approach presented here seeks to fill these gaps and provide near real time data on DOM conditions.

To date, most remote sensing of CDOM has focused on CDOM absorption. The two general algorithmic approaches to derive CDOM absorption are empirical approaches that relate in situ CDOM measurements with the remote sensing signal using statistical or machine learning techniques, and semi-analytical algorithms based on

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modeling the interaction of the light with the optical properties of the water (Cao et al., 2018; Griffin et al., 2018; Kutser, 2012; Lee et al., 2002; Wei et al., 2019). Both empirical and semi-analytical approaches have been used to derive CDOM absorption coefficients for inland and coastal water bodies with varying degrees of success (Al-Kharusi et al., 2020; Chen et al., 2019; Li et al., 2017; Olmanson et al., 2016; Pahlevan et al., 2022; Sun et al., 2021; Wei et al., 2019). Empirical models are developed with field data from specific water bodies within a unique timeframe and may not perform as well over a range of conditions and locations (Malthus & Dekker, 1995). Semi-analytical approaches, built on physics-based models, still also require field data for simulations or parameterization of model terms, and may not perform well at local or regional scales without location-specific tuning (Pahlevan et al., 2022; Ramsey, 2008). Both approaches are often limited by small data sets that result in bespoke models for a specific water body.

To address these challenges, we integrated satellite remote sensing and a large, continuously operating in situ optical sensor network that exploits the complementary advantages of both approaches to monitoring aquatic DOM. There is a burgeoning body of literature on the use of fDOM observations to develop remote sensing CDOM models (Peterson et al., 2020; Sagan et al., 2020; Slonecker et al., 2016). Yet, despite recommendations for more studies on the use of fDOM in remote sensing algorithm development given the ease of use of these in situ sensors (Slonecker et al., 2016), very few studies have made progress on this topic. In this study, we used an extensive data set of fDOM measurements from a sensor network of continuous in situ optical instruments distributed across a large freshwater estuarine complex. We demonstrated the utility of using fDOM for estimating DOM using remote sensing with high accuracy across a range of environmental and tidal conditions and seasons using machine learning. In addition, this work serves as proof-of-concept for development of an integrated sensor network that provides iteratively updated remote sensing retrieval models using near real-time sensor measurements and remote sensing imagery.

2. Study Area

We focused on the San Francisco Bay-Delta, the largest tidal estuary in the western Pacific, where the U.S. Geological Survey (USGS) maintains permanent continuous monitoring stations instrumented with fDOM sensors (EXO2, Xylem Inc., Yellow Springs, OH) alongside a suite of other optical water quality instruments. We focused our study on the freshwater portion of the estuary known as the Sacramento—San Joaquin River Delta (hereafter referred to as Delta), formed by the confluence of the two aforementioned rivers (Figure 1).

The Delta is a vital water resource for California, with almost half of the surface water in the state originating in the Delta watershed–draining a watershed of 153,000 square kilometers (km^2)–and flowing through or pumped into a system of federal and state water projects for water exports to agricultural and urban water users. The Delta provides drinking water for over 27 million people, water to support California's $54 billion agricultural industry (Delta Stewardship Council), and vital habitat for multiple endemic, threatened, and endangered species, including the protected delta smelt (US EPA, 2025). The Delta is managed by the California state agency, the Delta Stewardship Council, which created and implements the Delta management plan. The goals of the management plan are to provide a reliable water supply while protecting the Delta ecosystem (Delta Stewardship Council, 2025). Meeting the dual goals of water supply and ecosystem health requires heavy manipulation of water flow through the Delta to balance water use needs with water quality and flow targets for habitat protection. The majority of flow enters the Delta in the north via the Sacramento River; water pumps for exports are located in the southern Delta. During the dry summer months water is diverted from the Sacramento River in the north through the central Delta, controlled by gates in the Delta Cross Channel (Figure 1) (Monsen et al., 2007).

The Delta waterways are a network of riverine channels surrounding drained and leveed agricultural fields, tidal lakes, and marsh area. The region is characterized by wet winters and dry summer and fall seasons, with an average of 40% of the flow from winter runoff (Cloern et al., 2011). Daily median flows from the Sacramento River, the larger of the two rivers, vary seasonally from 320 to 1,230 cubic meters per second ($m^3\ s^{-1}$). The San Joaquin River daily median flows are 36–150 $m^3\ s^{-1}$ (CA DWR, 2023, median based on data from water years 1956–2001). The Delta is a tidal system, with complex salinity gradients that vary both with tidal and freshwater inputs on the scale of hourly to daily, as well as seasonally (Rath et al., 2017). Specific conductance values measured from 1965 to 2012 ranged from 0.1 to greater than 10 milliSiemens per centimeter (mS/cm) in the lower Sacramento and San Joaquin Rivers (Hutton et al., 2016). Sources for CDOM in the Delta vary in composition

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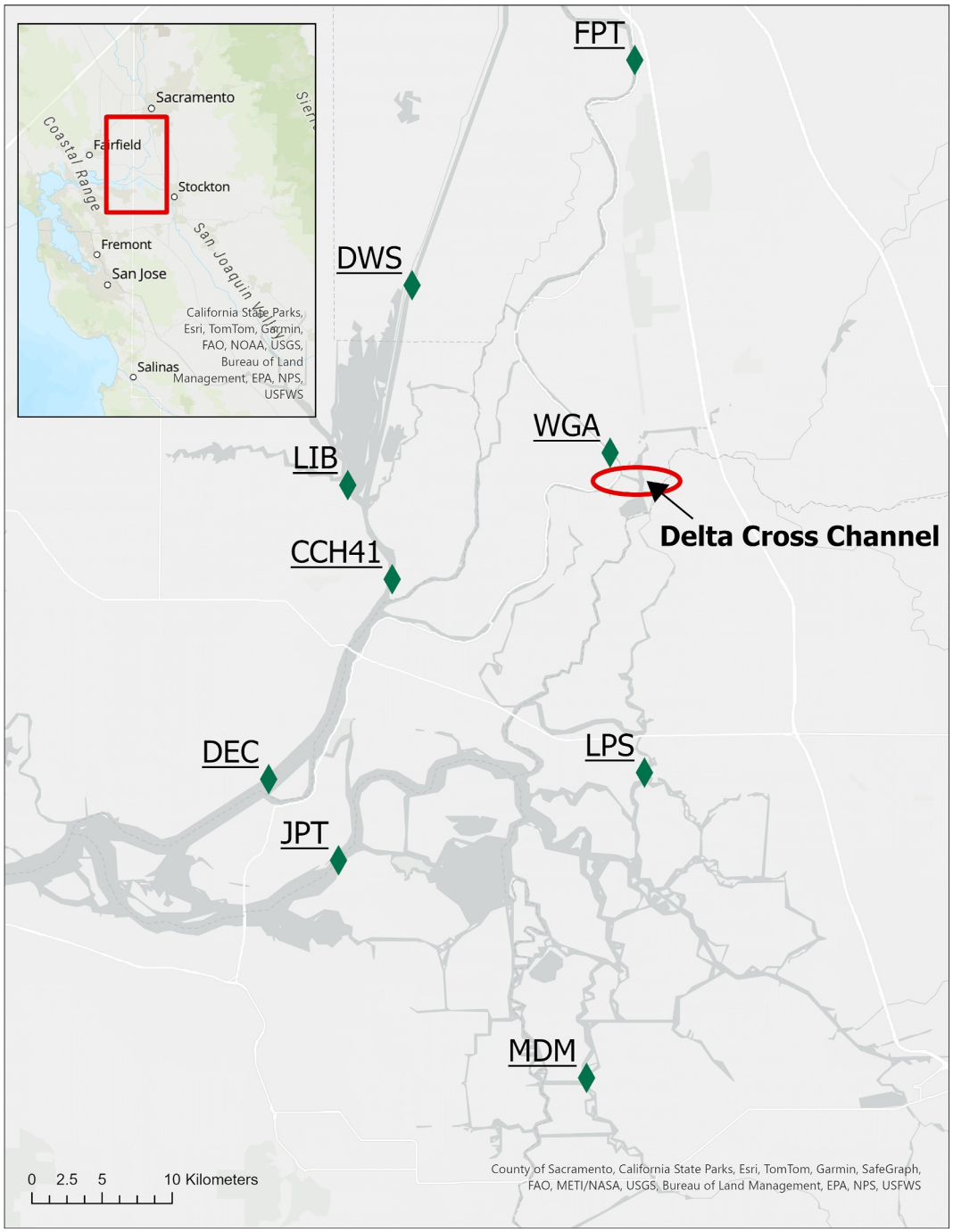

Figure 1. Study area map of the Sacramento—San Joaquin River Delta, California. Green diamonds indicate the locations of U.S. Geological Survey continuous monitoring stations where in situ fluorescent dissolved organic matter (fDOM) measurements were taken. The stations with fDOM sensors are: Freeport (FPT); the deepwater ship channel (DWS); Walnut Grove (WGA); Liberty Island (LIB); Ryer Island (CCH41); Little Potato Slough (LPS); Decker Island (DEC); Jersey Point (JPT); and Middle River (MDM). The red oval indicates the location of the Delta Cross Channel, which is used to control water flow during dry months.

seasonally. During the winter and spring months (December–April) DOM is mobilized from shallow, degraded peat layers by increasing groundwater levels, while during the summer and fall months (May–November) CDOM is sourced from the permanently saturated zone by groundwater flows (Deverel et al., 2007).

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Table 1
Fluorescent Dissolved Organic Matter (fDOM) Retrieval Model Calibration and Validation Data Sets Included in This Study of the Sacramento—San Joaquin River Delta, California

Data set	Location
Development, validation, and calibration of fDOM based remote sensing retrieval model	
Sentinel-2 Multispectral Imager (MSI) data	Continuous monitoring stations (2018–2020)
fDOM measurements by optical sondes	Continuous monitoring stations (2018–2020)
Independent fDOM based remote sensing retrieval model verification	
fDOM measurements by optical sondes	Ship based (2019–2021)
Historic CDOM retrieval model development and comparison with recent data	
Absorption and attenuation measurements	Historic data collected in the Delta (2014–2015)
Absorption and attenuation measurements	Ship based (2019–2021)

Note. CDOM = colored dissolved organic matter.

3. Materials and Methods

We used three field measurement data sets and one remote sensing imagery for this study (Table 1).

The availability of these field data sets combined with Sentinel-2A Multispectral Imager (MSI) remote sensing data facilitated the development of an fDOM based retrieval model. The ship-based field data sets of fDOM and spectral absorption and attenuation measurements provided the additional independent data needed to validate the model over a greater range of conditions. The historic data set provided context for our retrieval model.

3.1. In Situ Measurements

We used two in situ primary data sets in this study.

1. fDOM measurements from optical water quality sondes installed at nine different continuous monitoring stations (CMS) distributed throughout the Delta (Figure 1, Table 2) that made measurements every 15 min and thus provided concurrent match-up data with a satellite overpass (±15 min). These data were used to calibrate and validate the fDOM retrieval model used in this study.
2. Ship-based flow through measurements of fDOM and spectral absorption from an in situ spectral absorption and attenuation sensor (ac-s; Sea-Bird Scientific, Bellevue, Washington) that provided match-up data within

Table 2
Primary Data Set Measured at Nine U.S.

Station name	Latitude	Longitude	Number of samples	Range of fDOM (QSE)	Range of specific conductivity (mS/cm)	Range of turbidity (FNU)
CCH41	38.19396	−121.658	115	14.2–67.2	107.2–459.2	2.2–162.1
DEC	38.09333	−121.736	108	2.8–56.8	117.8–4684.6	3.5–137.1
DWS	38.34167	−121.644	122	14.4–46	283.8–730.6	4.7–65.4
FPT	38.45611	−121.5	126	8–53.4	84–231.2	1.4–61.2
JPT	38.05222	−121.688	126	19–84.4	116–2659.2	1.9–27.2
LIB	38.24222	−121.686	118	16–67.4	109.2–369.4	2–176.4
LPS	38.09639	−121.496	76	18–80	97.8–252.6	0.65–8.3
MDM	37.94277	−121.533	73	20.6–80.8	156.6–689.6	0.68–10.8
WGA	38.25778	−121.517	118	9–66	91.8–234.6	0.83–62.8

Note. Geological Survey continuous monitoring stations in the Sacramento—San Joaquin River Delta, California from 1 January 2018 to 31 December 2020. Station name, location, number of in situ measurements collected within ±15 min of a Sentinel-2 satellite image in the database, range of concentrations (minimum–maximum) of fluorescent dissolved organic matter (fDOM) in Quinine Sulfate Equivalents (QSE), specific conductivity in milliSiemens per centimeter (mS/cm), and turbidity in Formazin Nephelometric Units (FNU) are described for each station. Station names are: Freeport (FPT); the deepwater ship channel (DWS); Walnut Grove (WGA); Liberty Island (LIB); Ryer Island (CCH41); Little Potato Slough (LPS); Decker Island (DEC); Jersey Point (JPT); and Middle River (MDM). Data summarized from U. S. Geological Survey (2021) and Fleck et al. (2023).

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±3 days of a satellite overpass. The fDOM data from the ship surveys were used as a secondary independent validation data set for the fDOM model to test its extensibility to different locations within the study area, different dates, and different fDOM validation sensors. In order to compare the fDOM model performance with aCDOM model performance, the spectral absorption and attenuation data from the ac-s instrument were used to test a previously published aCDOM retrieval algorithm developed for Delta waters (Chen et al., 2019).

3.1.1. Continuous Monitoring Stations

The USGS maintains an extensive network of CMS instrumented with sensors to measure water quality in the Delta, that are routinely serviced on approximately a monthly basis. Nine of these stations are instrumented with EXO2 (Xylem Inc., Yellow Springs, OH) multiparameter water quality sondes carrying temperature, specific conductance (SC), turbidity, pH, dissolved oxygen, chlorophyll-a fluorescence, and fDOM probes (Figure 1, Table 2). The fDOM sensor emits UV light at 365 ± 5 nanometers (nm) and detects the DOM fluorescence response at 480 ± 40 nm (Booth et al., 2023). Following USGS protocols, each EXO2 sensor was calibrated in the lab to the Quinine Sulfate Equivalent (QSE) standard before installation (Booth et al., 2023). Each measurement was quality screened by applying a drift and fouling correction, according to Wagner et al. (2006), then corrected for temperature (rho = −0.01) (Watras et al., 2011) and turbidity (alpha = −0.003) (Booth et al., 2023), before being deposited in the USGS National Water Inventory System (NWIS) data repository. The data used in this study were downloaded from NWIS (U.S. Geological Survey, 2021) with a query for all valid measurements within ±15 min of a Sentinel-2A or -2B satellite overpass (Table 2) for the study period 1 January 2018 to 31 December 2020, and combined with data published in Fleck et al. (2023). This resulted in a total of 982 match-up data points over the study period. The size of this data set, with nearly concurrent match-up times, underscores the value of continuous monitoring stations maintained by the USGS. The number of samples from each CMS station location was reasonably evenly distributed across station location and season (Table S1 in Supporting Information S2). Variability in total number of measurements at these sites is due to routine maintenance schedules, and sites LPS and MDM were not installed until 2019.

3.1.2. Ship-Based Flow-Through Surveys

The USGS conducts regular ship-based mapping surveys using underway flow-through instrument systems. The surveys cover approximately 560 km of the Delta over three to four consecutive days, and makes a routine stop at all CMS stations in the Delta during the survey. The flow-through systems pass water pumped from approximately 1 meter (m) depth through a EXO2 multiparameter water quality sonde and two ac-s spectral absorption and attenuation sensors (10-cm pathlength). The EXO2 was equipped with temperature, SC, turbidity, pH, dissolved oxygen, chlorophyll-a fluorescence, and fDOM probes. For the spectral absorption and attenuation measurements, one ac-s instrument measured unfiltered total absorption and attenuation, the second passed water through a 0.2 μm pre-filter, measuring the dissolved fraction of the water. The ac-s instruments measured the absorption and attenuation coefficients in the range of 400–730 nm at a 4 nm resolution. The instrument was purged with deionized water at the end of each field day to confirm that drift was minimal. The data were smoothed using a cubic spline and resampled at a uniform 4 nm interval to match band positions for each ac-s instrument. The data were then corrected for salinity and temperature following the manufacturer recommendations using data collected from the flow-through system instruments described above. The data were then corrected for scatter using the Pope and Fry (1997) pure water absorption. In addition we used measurements of dissolved organic carbon (DOC) that were collected during ship-based surveys. These samples were analyzed by the U.S. Geological Survey Organic Matter Research Laboratory following the methods in Potter and Wimsatt (2009), and are available through NWIS (U.S. Geological Survey, 2021). The measurements collected during the routine stops at the CMS stations were used in this study (Fleck et al., 2023).

A total of four surveys were conducted for the study, one in each season (winter: January 2020; spring: April 2021; summer: July 2020; autumn: October 2019), resulting in a data set with 107 match-up data points for aCDOM (defined as the absorption coefficient at 440 nm (a440)) and 106 match-up data points for fDOM collected during the routine stop at a CMS that were within ±3 days of a Sentinel-2A or -2B satellite overpass. The surveys were designed so that the sampling was done within ±2 hr of tidal stage similar to when the satellite overpass occurred; the number of samples collected was consistent for each survey (Table S2 in Supporting Information S2).

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3.2. Satellite Data Acquisition and Pre-Processing

The European Space Agency's (ESA) Sentinel-2A and -2B (S-2A and S-2B, respectively) MSI data were used for this study. The frequent revisit time (5 days) and relatively high spatial resolution (20 m) of the data make it ideal for inland water bodies. The tile that encompasses the Delta is T10SFH. Each T10SFH tile with less than 25% cloud cover acquired from 1 January 2018 through 31 December 2020 was downloaded from the ESA Copernicus online database (Copernicus Data Space Ecosystem, 2023) as a Level 1C (L1C) product. Every image was also visually inspected to ensure that the Delta was cloud free. This yielded a database of 131 images. The MSI sensor is equipped with 13 bands in the visible, near-infrared and shortwave-infrared (VIS-NIR and SWIR, respectively). Bands 1–7 (VIS-NIR) and Band 11 (SWIR) were used in this study.

The open-source software package ACOLITE v20180327.0 (Vanhellemont, 2021) was used to atmospherically correct each S-2A/S-2B L1C T10SFH tile to remote sensing reflectance (Rrs) products. The dark spectrum fitting (DSF) method was implemented in the processing, as it has been shown to improve retrievals in the blue bands (Vanhellemont, 2019; Vanhellemont & Ruddick, 2018). Rrs was calculated for each image in the database, at a pixel resolution of 20 m. This pixel size is ideal for use in the Delta where the narrow width waterways are not spatially resolvable by sensors with a pixel resolution of greater than 30 m (Kuhn et al., 2019). ACOLITE has been shown to be reasonably accurate for other water quality studies in the Delta (Lee et al., 2021). Finally, a land mask was applied to each image using a threshold value of Rrs greater than 0.009999 in the SWIR band at approximately 1,610 nm (S-2A/S-2B Band 11). Water is highly absorbing in the short-wave infrared range (Curcio & Petty, 1951) while vegetation and soil have higher reflectance values in the infrared. All pixels with a value greater than 0.009999 in Band 11 were reassigned a value of NA.

For each image in the combined S-2A/S-2B database, the following steps were followed for extracting pixel values centered on the geolocation of each CMS and ship-based survey data collection point.

1. The recorded location of each in situ measurement was checked for accuracy and shifted to the nearest water pixels if it was identified as on land.
2. All combinations of band ratios for Rrs bands 2–5 (B2-B5; 492–704 nm) were calculated for each image.
3. The mean value in a 3 by 3 pixel window, centered on the geolocation of each CMS station, was extracted for each Rrs band (B2-B5) and band ratios.

These data were then matched to the CMS data collected within ±15 min of the satellite overpass ($n = 982$; Table 2), and the ship survey data (Table S2 in Supporting Information S2; $n = 106$ for fDOM and $n = 107$ for a440). The CMS and S-2A/S-2B data set forms the basis of the empirical algorithm development. The ship survey and S-2A/S-2B data set were used as independent validation of the empirical algorithm developed in this study and to validate the Chen et al. (2019) algorithm applied to the S-2A/S-2B data in this study (Text S1 in Supporting Information S1).

3.3. Empirical Algorithm Form

For this study, the fundamental shift in approach is using in situ fDOM measurements for model development using a machine learning approach underpinned by the physics of the measurements. S-2A and S-2B bands B2 (blue), B3 (green), and B4 (red) were included as input variables for the empirical model development. Band ratios that have been shown to be correlated with CDOM retrievals in other studies were also included as input variables. These include B2/B3 (Griffin et al., 2011; Liu et al., 2021), B3/B4 (Brezonik et al., 2015; Chen et al., 2019; Herrault et al., 2016), and B4/B2 (Alcântara et al., 2016; Xu et al., 2018). In addition, season was included as a categorical variable to capture the seasonal variability in CDOM character and distribution in the Delta.

3.4. Model Calibration and Validation

We used a machine learning approach to develop an fDOM retrieval model from the CMS and S-2A/S-2B data using the CMS in situ fDOM measurements as the response variable. All data handling, calculations and visualization were performed in R (R Core Team, 2021), and all machine learning was implemented in R using the caret package (Kuhn, 2008). The default setting in the caret packages chooses the best regression model based on the lowest mean squared error (MSE). This model is identified in the output.

The match-up data were subset into independent calibration and validation data sets using a random split of 70% calibration ($n = 689$) and 30% validation ($n = 293$) subsets. This random split was bootstrapped 1,000 times with replacement to characterize the effect of the distribution of the calibration data set on model performance following Bolch et al. (2020). Each calibration data set was used as an input for four modeling frameworks: random forest (RF), stepwise forward multiple linear regression (FMLR), partial least squares regression (PLS), and non-linear least squares regression (NLS). For each of the 1,000 model runs, a 10-fold cross validation was applied to the calibration data set for evaluation of model performance. The best performing model of the 10-fold cross validation was identified for each calibration data set by using the lowest MSE as the evaluation criteria. The best performing model for each of the 1,000-run model framework was then applied to the independent validation data set for that model run. Calibration model performance and independent validation accuracy statistics for each of the 1,000 model runs was compiled.

3.4.1. Model Performance

For each of the 1,000 model runs, the R^2, root mean square error (RMSE), normalized RMSE (nRMSE), and mean relative error (mRE) were calculated from the independent validation data. The best performing model (based on RMSE and R^2) was chosen from the four modeling frameworks to apply to the time series of Sentinel-2 imagery for visualization of fDOM retrievals in the Delta and subsequent analysis of model performance.

To evaluate the environmental effects on fDOM model performance, we assessed the performance of the model across site and season by adding these variables as interactions in the regression model between model predictions and the independent validation data. We then assessed water quality conditions by independently and then sequentially adding turbidity, chlorophyll-a fluorescence, and specific conductivity as interaction terms in the regression model.

Finally, we also applied the best performing model to the extracted ship survey data to perform one additional, independent validation using the secondary validation data set. This allowed us to quantify the model extensibility to different locations within the study area across a data set with much less restrictive match-up times and applied to a different set of fDOM sensors (same manufacturer).

4. Results

4.1. Machine Learning Model Results

Both the RF model and the FMLR model consistently performed best (Figure S1 in Supporting Information S1) when applied to the independent validation data. The non-linear least squares models consistently performed poorly. Performance across the 1,000 runs of each model framework had variability across all modeling frameworks; but RF and FMLR had the lowest amount of variability between runs, whereas PLS and NLS had the greatest variability.

Variable importance is a measure of the contribution of each input variable to the model and provides insight into the predictive power of each variable. Variable importance is calculated by assessing the change in prediction accuracy before and after each predictor value is added to the model, then averaged over all trees in the model and normalized by the standard error (Kuhn, 2008). This was calculated for each RF model run as a standard output metric by the R caret packet used for model calibration and validation. For the 1,000 RF model runs, season and the ratio of the red to blue band (B4/B2) were the two highest ranked variables in every iteration of the RF model run. The red band (B4) notably was included in none of the models, while the other variables (blue band, green band, and band ratios) were included with varying importance.

The best performing RF model (R^2 = 0.76, RMSE = 6.1 QSE, Figure 2) was chosen for discussion for the purposes of this paper. The two variables with the highest importance in this model are the B4/B2 (100 importance); and season (77.5 importance). Also included are the green band (67.1 importance), the blue band (51.3 importance), the ratio B3/B4 (35.2 importance), and ratio B2/B3 (29.4 importance). This model slightly overpredicts at lower fDOM values and slightly underpredicts at higher values. This may be due to the distribution of the data, where most values are less than 40 QSE.

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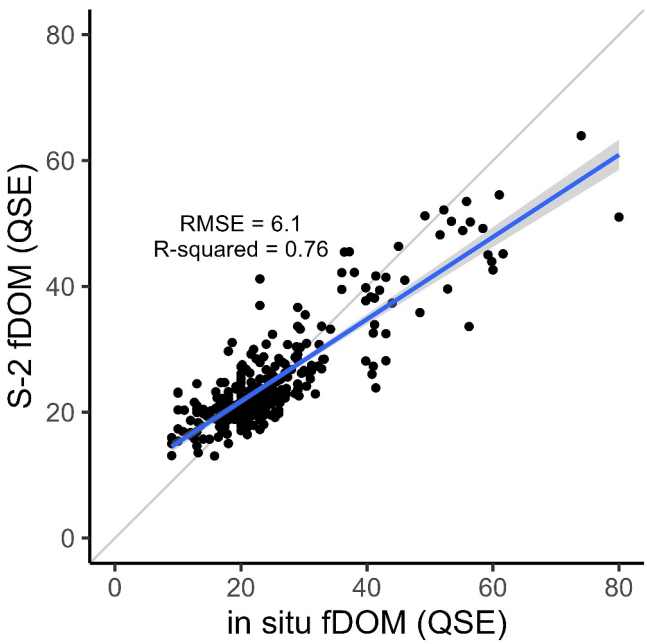

Figure 2. Fluorescent dissolved organic matter (fDOM) in Quinine Sulfate Equivalent (QSE) predicted by the S-2A/S-2B random forest model, compared to in situ fDOM measurements from U.S. Geological Survey continuous monitoring stations in the Sacramento—San Joaquin River Delta, California. RMSE = root mean squared error.

4.1.1. fDOM Model Performance Across Season, Site, and Water Quality Conditions

Higher fDOM values typically occur in the Delta during the winter (1 December–28 February) and spring (1 March–31 May) months, with the lowest values in the summer (1 June–31 August) and fall (1 September–30 November) months. Season was included as a categorical variable in the model inputs to account for this variability. The RF model presented here performed well across all seasons with the best performance for winter and spring (R^2 = 0.67, RMSE = 9.0 QSE (winter), 8.7 QSE (spring)). The RF model performed reasonably for summer (R^2 = 0.54, RMSE = 4.2 QSE), and fall (R^2 = 0.43, RMSE = 4.0).

fDOM varies spatially (Table 2) throughout the Delta, and thus model performance varied by station. The model performed well across all sites, with $R^2 \geq 0.72$ for all sites except DWS (R^2 = 0.66) and WGA (R^2 = 0.53), indicating that our model represented most sites quite well, with the lowest R^2 values at sampling locations in the narrowest of channels. When site was added as a variable in the RF models, the variable importance of any site was consistently less than 20%, indicating that site location did not have a large impact on the overall performance of this model.

In addition to the temporal and spatial variability of fDOM, the impact of turbidity, SC, and chlorophyll-a fluorescence on the RF model performance was assessed (Figure S2 in Supporting Information S1). When each variable was added to the RF model, the model performance did not improve or decrease. In addition, the scarcity of chlorophyll-a fluorescence measurements reduced our sample size by 45%. Because these input variables had little impact on the model, the initial remote sensing only model was chosen. The impact of turbidity, SC, and chlorophyll-a were also assessed by adding those variables as an interaction term in the linear regression between in situ and predicted fDOM. The interaction of modeled fDOM and SC was significant at $p = 0.055$, but no other interaction terms were significant. This result supports the minimal impact on the random forest model when those measurements were added as variables.

4.2. fDOM Model Applied to Ship Survey Data

Validation results for the fDOM model applied to ship survey data as a second, independent validation data set yielded an R^2 = 0.48, RMSE = 12.5 QSE, nRMSE = 72.1, and mRE = 27.44 (Table 3). While model performance applied to this validation data set was not as high as the validation with the CMS validation data set, the range of fDOM in the ship survey database was different (9.6–86 QSE, Table S2 in Supporting Information S2), and did not include very low fDOM values (<9.5 QSE). This may contribute to the lower R^2 and higher RMSE results

Table 3
Comparisons Between Validation Results for the Chen Colored Dissolved Organic Matter Spectral Absorption Coefficient (aCDOM) at 440 nm (a440) Model (Chen et al., 2019) and the Random Forest (RF) Florescent Dissolved Organic Matter (fDOM) Model in Quinine Sulfate Units (QSE) in the Sacramento—San Joaquin River Delta, California

Model/Validation	R^2	RMSE	nRMSE	NSE	mRE	n
RF fDOM	0.76	6.1 QSE	51.9	0.50	17	982
RF fDOM ~ ship based fDOM (secondary validation)	0.48	12.5 QSE	72.1	0.47	27	105
S-2 Chen ~ ship based ac-s (a440)	0.40	0.34 per meter	90	0.18	37	107
Chen (Landsat 8)	0.78	0.161 per meter			26	66

Note. The ship-based fDOM is the secondary independent validation data set (Fleck et al., 2023). The Sentinel-2 (S-2) Chen model is from Chen et al. (2019) applied to S-2 imagery on dates when the ship-based absorption measurements were taken. RMSE, root mean square error; nRMSE, normalized RMSE; NSE, Nash Sutcliff Efficiency; mRE, mean relative error; n, number of samples, ~ = regression.

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because the RF model incorporates a larger range of fDOM values. Additionally, the time gap between the ship survey data and the S-2A and S-2B imagery is ±3 days, a much larger time gap than the ±15 min for the CMS data set used in model development. This time gap can also lead to lower R^2 and higher RMSE because conditions can change quickly in the Delta, highlighting the need for concurrent measurements.

4.3. Comparison With Chen aCDOM Model

We applied the Chen aCDOM model (Chen et al., 2019; Text S2 in Supporting Information S1) to our S-2A and S-2B Rrs data set acquired within ±3 days of the ship-based ac-s data set (a440; $n = 107$). The validation results for the Chen aCDOM yielded an $R^2 = 0.4$, RMSE = 0.33, nRMSE = 90, and mRE = 37.0 (Table 3). The results of the Chen aCDOM model are highly similar to the validation results of our fDOM model with the same ship survey data set, but much lower than those published by Chen et al. (2019) using the historical data.

5. Discussion

This study explored the use of integrated sensor network measurements for development of a remote sensing fDOM retrieval model by coupling near concurrent (±15 min) S-2 reflectance and in situ fDOM measurements from field-based sensors that were deployed in a monitoring network throughout the Delta. Machine learning techniques were used to develop a robust model for satellite-based fDOM retrieval models. This study used an extensive set of match-up data between fDOM measurements from nine continuous monitoring stations and S-2 remote sensing observations ($n = 982$), ranging across hydrological conditions, tidal conditions, seasons, and years (2018–2020) to develop and validate a remote sensing algorithm for fDOM, one of only a few in the literature to date. We further validated this algorithm on a completely independent set of fDOM data collected from ship-based surveys to test the extensibility across space and sensors (of the same manufacturer), and then compared these results to a previously published aCDOM algorithm developed for the Delta (Chen et al., 2019). The fDOM retrieval results from the top performing models were consistent with those published previously in the literature, and to those from Chen et al. (2019). Retrieval of fDOM maps from S-2 imagery provided a time series that is easily comparable to and complements readily available, relatively low-cost in situ sensors, enabling exploration of spatial and temporal patterns of DOM, which may be useful for biogeochemistry and ecology studies.

5.1. Model Performance

The exceptionally large size of this data set and machine learning-based approach to developing a remote sensing fDOM retrieval model allowed assessment of both spatial and temporal performance and suggests that an integrated sensor network is highly advantageous for robust remote sensing retrievals. While previous studies have been successful to varying degrees in developing CDOM retrieval models, the typical sample size is small. Thus, studies usually are not able to conduct truly independent validation and have limited statistical power to explore the effect of environmental conditions or test the full extensibility of the approach across tidal stages, seasons, locations, or instruments. This is often due to financial and time constraints on field work. For example, Kutser (2012) had a sample size of 15 over one lake in Sweden. Brezonik et al. (2015) used a data set of 34 samples in Minnesota Lakes and S-2 imagery collected with a window of 1–2 months of the in situ samples. Alcântara et al. (2009) used a data set of 32 samples collected in May and October 2014 in a Brazilian reservoir and developed a CDOM retrieval algorithm using above water radiometry measurements. This algorithm was then applied to Landsat 8 imagery. Twenty-three samples collected between August 3 and 8, 2015 from Estonian lakes and three S-2 images were used by Toming et al. (2016) to test the performance of S-2 imagery. The database used by Chen et al. (2019) was larger than many studies, comprised of 66 match-up points collected over 4 days in the Delta and Lake Erie. The size of the data set used in this study for fDOM model development is approximately 200 times larger than previous studies. This is due to the temporally dense in situ data from USGS CMS and more frequent combined overpass occurrence of S-2A and -2B satellites (5 days). The performance of this fDOM model suggests that integrating a network of in situ water quality sensors with concurrent remote sensing imagery into a flexible model framework could improve retrievals of water quality parameters and provide a robust quantification of the uncertainty under varying and often rapidly changing environmental conditions typical of tidal environments.

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The most accurate fDOM retrieval models in this study came from the RF and FMLR frameworks, where maximum R^2 values for validation were 0.76 (RF) and 0.67 (FMLR), and minimum RMSE values for validation were 5.4 QSE (RF) and 6.4 QSE (FMLR) (Figure S1 in Supporting Information S1). These results are consistent with the range of model performance in other studies when validation metrics are published. Alcântara et al. (2016) reported a validation $R^2 = 0.7$ and RMSE of 10.65% for the empirical model based on a440 measured in Brazilian reservoirs using the S-2 B4/B2 band ratio. Likewise, Xu et al. (2018) reported a validation $R^2 = 0.7$ for the empirical model based on a355 in Poyang Lake, China using Landsat 8 band ratios, RMSE was not reported for this study. Results from model calibration are more commonly reported in current literature than independent validation. Recent studies by Slonecker et al. (2016), Sagan et al. (2020), and Peterson et al. (2020) present retrieval algorithms based on fDOM measurements. Slonecker et al. (2016) reports a calibration $R^2 = 0.418$ for the algorithm using Landsat 8 bands. Typically, these retrieval algorithms are empirical in nature and are based on a regression model where bands and band ratios are input variables. However, machine learning techniques are gaining traction for DOM retrieval models in inland and coastal waters. The support vector regression model developed by Sagan et al. (2020) using 83 fDOM measurements from Illinois lakes and the National Aeronautics and Space Administration (NASA) Harmonized Landsat and Sentinel-2 (HLS) product had a calibration $R^2 = 0.59$. Peterson et al. (2020) developed a neural network retrieval model using 97 match-ups with fDOM measurements from Illinois lakes and the NASA HLS, with model calibration results of $R^2 = 0.92$ and RMSE = 14.4 QSE. While the Peterson et al. (2020) model used S-2 data as model inputs, the resulting neural network model was not available to test. The large data set used in this study facilitated the development of the RF model presented here in addition to primary (CMS measurements) and secondary (ship-based measurements) validation processes.

The success of multiple machine learning frameworks (RF and FMLR) suggests that models built on integrated sensor networks are relatively model-framework agnostic for regional areas. However, the relatively large range of model performances within a given framework based on permutations of training and test data splits indicate that each of these frameworks are sensitive to the training and test data. This suggests that while machine learning frameworks are well suited to nonparametric regression (Lary et al., 2016), developing multivariate nonlinear relationships, avoiding overfitting and unbalanced data sets, and identifying how important input variables are to the model (Belgiu & Drăgu, 2016; Lary et al., 2016) continue to pose challenges for model development (Wang & Wang, 2020). This challenge can be met by further expanding data sets with concurrent in situ and remote sensing observations that represent a broad range of environmental conditions. This study suggests that DOM retrieval models, particularly in a dynamic and highly variable system such as the Delta, would benefit from using large data sets that include a range of DOM conditions and can be tightly temporally coupled with remote sensing image acquisition.

Examining variable importance metrics for the RF model can lead to insights about the physical processes driving a successful fDOM remote sensing model. The top three variables in the RF models were the ratio of the red to blue bands (B4/B2), season, and the green band (B3). The high importance of season reflects the variability of CDOM in the Delta, where CDOM is higher in winter and spring and has a different character than in summer and fall. In addition, the S-2 bands included in the model are frequently identified in empirical algorithms that are developed with both absorption and fluorescence measurements of DOM (i.e., Chen et al., 2019; Brezonik et al., 2015, and references therein). The green band captures the dominant color of water reflectance and encompasses wavelengths where CDOM absorption is still a contributing factor (Kutser et al., 2005). Short-to-long wavelength ratios are used to capture higher contribution of CDOM absorption (short wavelength band) while removing the effects of particulate matter (Chen et al., 2019) and chlorophyll (Brezonik et al., 2015). In this study, the sensors that measure fDOM rely on excitation-emission wavelength pairs to measure DOM. The sensor contains a light that emits energy at a known wavelength in the UV portion of the electromagnetic spectrum, and then records the amplitude of the fluorescence response at a higher wavelength in the visible range (Booth et al., 2023). Remote sensing instruments with bands in the range of the fDOM sensor record the naturally induced fluorescence emission, where UV light from the sun is the source of energy at excitation wavelengths. Therefore, it is unsurprising that the Rrs bands and band ratios that were most important in an empirical model for fDOM retrievals were similar to the emission bands in the in situ fDOM sensors deployed in the field. In this case, the EXO2 sensor emits UV light at 365 ± 5 nm and detects the DOM fluorescence response at 480±40 nm, so the blue and green S-2 bands were the most likely to be included in our model. The red band performs the same function as it does for absorption algorithms.

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The top RF fDOM retrieval model used for subsequent environmental analysis performed well overall but performed better in the winter and spring seasons. This may be due to low DOM in the summer and fall, combined with water management operations. During the summer months (June–August) the Delta Cross Channel, located near site WGA (Figure 1) is opened to ensure sufficient water supply to the drinking water pumps in the southern Delta. This action results in a more complicated hydrologic system within the Delta, where the Sacramento River becomes the dominant water source throughout the central and western part of the Delta in the summer months (Monsen et al., 2007). The result is a source of water that is more optically uniform moving throughout the Delta. Four of the seven sites (DWS, LIB, CCH41, and DEC) are located in the western part of the Delta, where lower flow in the summer results in slower movement of water and encroachment of brackish water from the estuarine reaches of the Delta, potentially impacting model performance for that season. Additionally, summer drainage of agricultural islands in the Delta contributes smaller amounts of DOM derived from groundwater movement through deep saturated peat rather than the large amounts of DOM drained from the degraded peat deposits during winter and spring (Deverel et al., 2007; Richardson et al., 2020). During winter, the degraded peat DOM mixes with multiple riverine inputs, including Sacramento River and San Joaquin River sources, as well as Cosumnes and Mokelumne Rivers containing terrestrially-derived DOM draining from the extensive agricultural lands of California's Central Valley. This results in DOM with different spectral signatures in winter, particularly at the southern sites, MDM and LPS, which can be dominated by water from the San Joaquin River and the Cosumnes and Mokelumne Rivers, respectively, following winter and spring rains.

The consistent performance of the model at all sites indicates that the spatial and temporal distribution of samples is robust, and a single model for fDOM retrievals is sufficient to describe fDOM variability across the Delta. The two sites with the lowest model performance were also in the narrowest channels in the study. Site DWS is located in the Sacramento Deep Water Ship Channel, a levee-bounded channel maintained by the U.S. Army Corps of Engineers for freighter traffic to the Port of Sacramento. Similarly, site WGA is located in a narrower channel of the Delta. The relatively narrow channels and the optically bright levee roads cause adjacency effects in the water pixels of the channel. This adjacency effect is visible in the unusual brightness of the blue and green bands in the pixel reflectance spectra at the DWS site. High green band (559 nm) Rrs values due to factors other than levels of DOM in water will impact model performance because the green band is an important variable in this model. To minimize this issue, areas where likelihood of adjacency effects are high (i.e., narrower channels, or where levee constructions are optically bright) could be excluded from the data set or masked from future analyses. As adjacency corrections continue to improve (Paulino et al., 2022) this issue may be resolved with application of a robust correction to the remote sensing data.

5.2. Spatio-Temporal Patterns in DOM Distribution Across the Delta

The complexity of DOM in the Delta is due to tidal, seasonal, and spatial variability of input sources. Delta DOM is comprised of inputs from the Sacramento and San Joaquin Rivers, as well as downstream inputs from wetlands and drained island sources. Overall, the composition and source of DOM is primarily from the river inputs, but the DOM from wetlands and drained islands contribute 13%–49% (Kraus et al., 2008) of the dissolved organic carbon (DOC) exported from the Delta. This variability is further compounded by interannual variability due to wet/dry year cycles. Tidal flow also plays a role in DOM export from wetlands, modified by connectivity to river flow, with higher DOM export in summer, fall, and winter (Lehman et al., 2015). Additionally, water is pumped out of drained islands into Delta waterways through pipes and is discharged from agricultural fields, forming point sources for DOM in the Delta that integrate into the waterways, with the greatest concentrations during the winter months (Eckard et al., 2020; Richardson et al., 2020). Eckard et al. (2020) found that the DOM signature from terrestrial inputs can be identified in the riverine DOM signature downstream. In wet years, river inputs are the dominant source of DOM, while in dry years sources from wetlands and drained islands that were converted to agricultural use contribute a larger portion of DOM (Jassby & Cloern, 2000). During our study period, 2019 was a wet water year, while 2020 and 2021 were exceptionally dry water years. We would expect to see the input of DOM from wetlands and agricultural drained islands in the winter and spring of 2020 and 2021 during this study period. In 2019, we would expect the dominant DOM input source to be smaller riverine tributaries in the Delta.

We visualized the highest performing RF fDOM model applied to S-2A and S-2B images acquired on 9 February 2020 (winter), 5 May 2021 (spring), 23 July 2020 (summer), and 26 September 2021 (autumn) to characterize Delta-wide spatio-temporal patterns of DOM (Figure 3). The distribution of DOM in winter 2020 was highest in the middle of submerged islands (i.e., Liberty Island in the north and Franks Tract in the central Delta), with lower

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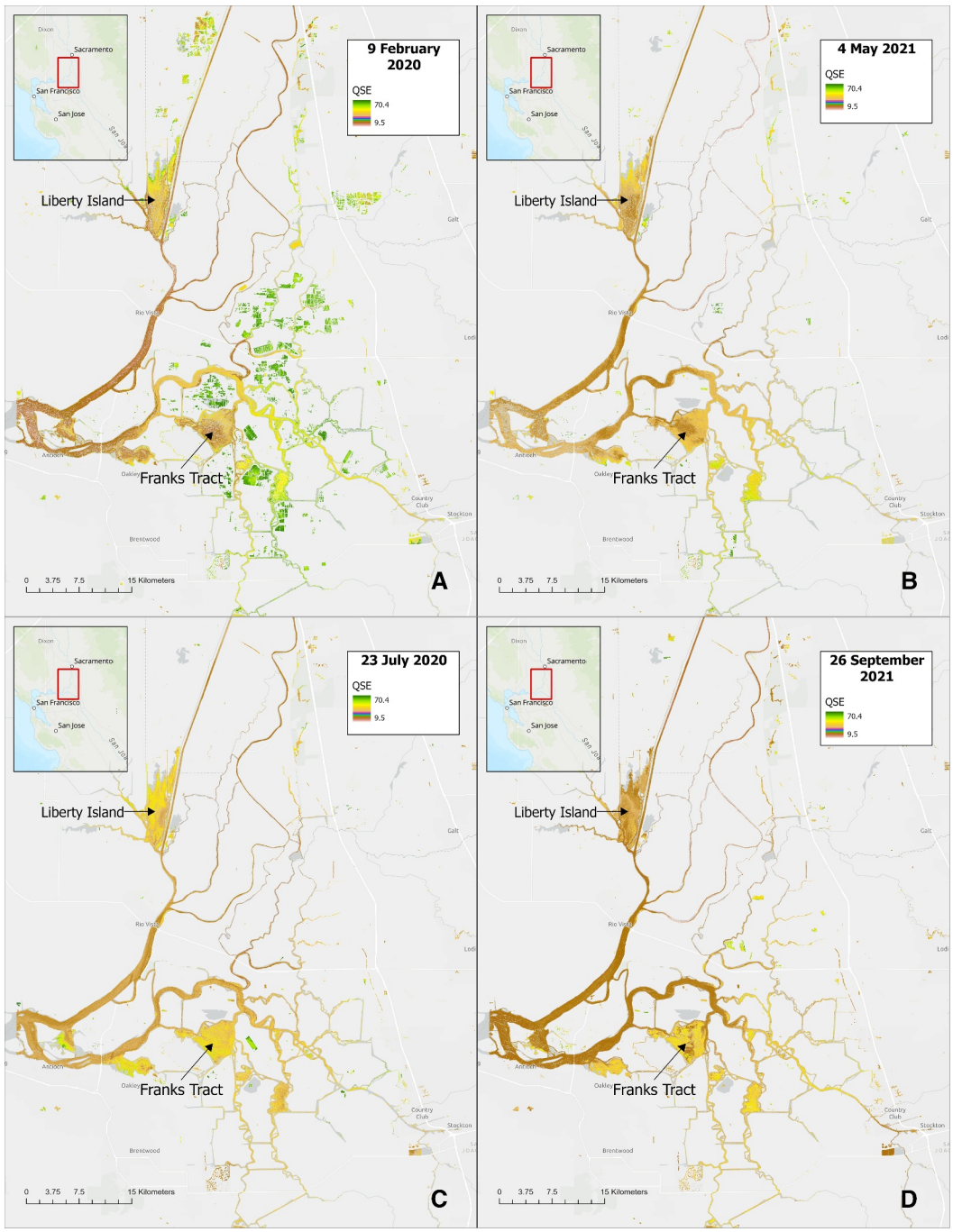

Figure 3. Map of fluorescent dissolved organic matter (fDOM) predicted by the Sentinel-2 random forest (RF) model in (a) winter (9 February 2020), (b) spring (4 May 2021), (c) summer (23 July 2020), and (d) autumn (26 September 2021). These specific dates were chosen during our study time range because the Sentinel-2 satellite imagery was cloud-free. The maps visualize the distribution of fDOM in the Sacramento—San Joaquin River Delta, California during different seasons. Basemap by Esri (2021).

amounts close to shore, and in the western river channels. High levels of DOM were present in flooded agricultural areas in the central Delta (green areas in Figure 3a). In spring 2021, DOM levels continued to be high in open areas, but highest close to shore and in the western river channels of the Sacramento and San Joaquin Rivers (Figure 3b). These patterns suggest that DOM from the flooded agricultural areas was discharged into Delta waterways in the three months between the winter 2020 image and the spring 2021 image. The autumn image

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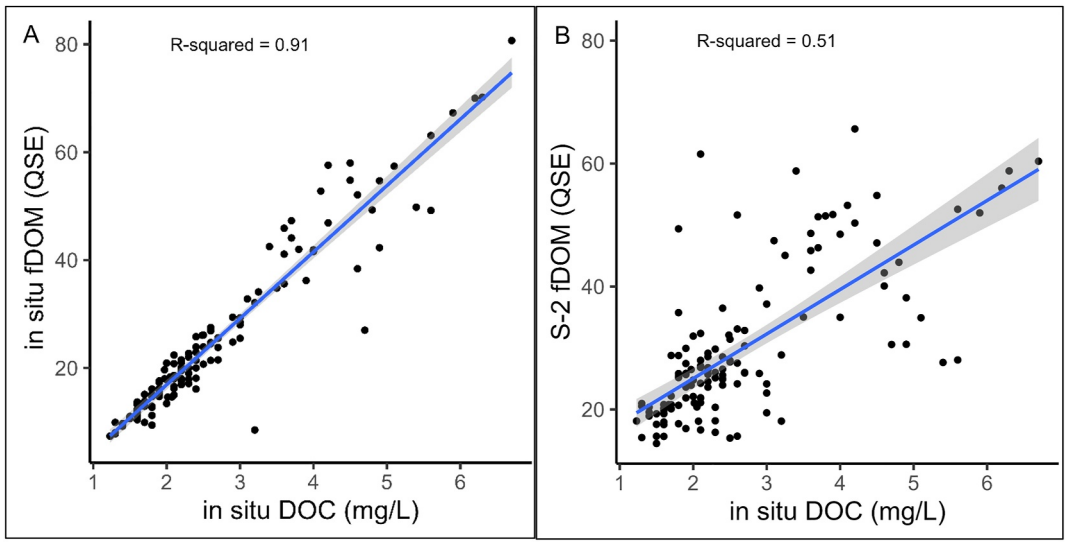

Figure 4. Relationship between fluorescent dissolved organic matter (fDOM) and dissolved organic carbon (DOC) in the Sacramento—San Joaquin River Delta, California. Panel (a) shows the relationship between in situ fDOM measurements in Quinine Sulfate Equivalent (QSE) and in situ DOC (mg/L) measurements collected during ship-based surveys (Fleck et al., 2023). Panel (b) shows the relationship between predicted fDOM from a random forest model using Sentinel-2 satellite imagery and in situ DOC measurements collected during ship-based surveys.

shown was acquired in September 2021, at the end of a very dry water year and in the beginning of the autumn season (Figure 3d). The overall range of DOM was lower in autumn. The highest levels of DOM were in open areas in the Delta, where the DOM signature was likely from reduced riverine inputs during the overall dry water year. Additionally, there were areas where agricultural islands, in the north and eastern areas of the Delta, were still flooded enough to show up in the satellite imagery. This suggests that the agricultural drainage is still a contribution to DOM levels in autumn 2021.

5.3. On the Utility and Reliability of an fDOM-Based DOM Product

We found that measurements of DOC and fDOM collected during the ship-based survey were linearly related ($R^2 = 0.91$) (Figure 4a), indicating that fDOM measurements are a good indicator of DOC levels in this system. DOC is linearly related to the RF fDOM model retrievals ($R^2 = 0.51$) (Figure 4b).

Similarly, Hernes et al. (2009) found that fDOM is a reliable indicator of DOM levels in the Delta. Furthermore, Nieke et al. (1997) found fDOM to be a reliable indicator of DOM in the St. Lawrence Estuary system. The close relationship of fDOM and DOC in the Delta is encouraging for the use of deployable sensors, and the utility of large in situ data sets that represent a full range of conditions for satellite retrieval model development. Our model shows very high performance for fDOM estimation when using independent CMS data reserved for validation and performs comparably to absorption-based methods for data collected within ±3 days of satellite overpass. These results suggest that implementing an integrated sensor network of water quality sensors could provide valuable measurements for better understanding complex systems.

5.4. Conclusions and Implications

This study explored the use of near concurrent (±15 min) satellite data and in situ fDOM measurements from sensors that were deployed in a network throughout the Delta. Machine learning techniques were used to develop a robust model for satellite-based fDOM retrieval models. The close relationship of fDOM and DOC in the Delta is encouraging for the use of deployable sensors, and the utility of large in situ data sets that represent a full range of conditions for satellite retrieval model development. Our model shows very high performance for fDOM estimation when using independent CMS data reserved for validation and performs comparably to absorption-based methods for data collected within ±3 days of satellite overpass. These results suggest that implementing an integrated network of water quality sensors provides valuable measurements for better understanding complex

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systems. The USGS continuous monitoring sensor network in the Delta is one example of how a network of sensors can be established and maintained.

The complexity of DOM in the Delta highlights the need for an integrated network that includes both remote sensing and in situ sensors. Remote sensing models provide information on monthly to seasonal timescales (when accounting for cloud cover) at Delta-wide spatial scales. In situ sensors provide point measurements on short (approximately 15 min) temporal scales that can resolve the effect of shorter temporal variables such as tidal cycles and storm events. This approach to water quality retrievals is best suited to water bodies where frequent monitoring via a distributed network of sensors is possible, which is necessary to characterize variability in the system and to provide data for management decisions. In situ water quality sensors provide high precision, temporally rapid data at the individual point level, while satellite-based remote sensing data provides system-wide coverage at a known revisit time. Integrating these systems provides comprehensive, large scale coverage.

This approach begins to mimic the operational climate model approach, where models are continuously updated as new observations are collected (Righi et al., 2020), and could be applied to other areas such as the Great Lakes where large-scale monitoring is required and is often already in place. In the United States there are buoy-based in situ sensors deployed in the coastal oceans and throughout the Great Lakes that provide continuous water quality measurements as part of the NOAA Integrated Ocean Observing System (NOAA, 2022) (https://ioos.noaa.gov/). The network of buoys in the Great Lakes are maintained by the Great Lakes Observing System (GLOS) partnership. The GLOS network of buoys monitor both water quality parameters and weather conditions at numerous locations throughout the Great Lakes, upload the data in real time for public use through an Internet of Things (IoT) based data platform (GLOS, 2023). The USGS is developing and expanding on existing continuous monitoring networks through the Next generation Water Observing System (NGWOS, 2021) by installing and maintaining integrated water quality sensors along with other hydrologic instrumentation to comprehensively monitor water body conditions. These efforts reflect barriers to implementation: namely, the cost of installation and upkeep, as well as the need for both human and computing resources for data quality assurance and quality control, data storage and access, and model updates as technology advances. These obstacles are currently being overcome by building networks of collaborative stakeholders to handle the challenges.

Developing an iterative automated model framework that accesses and integrates near real-time satellite data and in situ fDOM measurements on a weekly basis would likely improve the fidelity of the modeling and make for dynamic decision making in our rapidly changing environment. To achieve this would involve expanding computing capabilities and automating image processing and modeling capabilities within the continuous monitoring data computation infrastructure. Additionally, further work focusing on the variability in the Delta that drives the poor performance of the summer season in this model could improve our understanding of the factors affecting water quality during this biologically important period of the year. This may involve developing different models for fDOM retrievals that are applied during different seasonal, tidal, or management action time periods in order to better capture spatial and temporal DOM distribution.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data used in this study are freely available. The Sentinel-2 Level 1C remote sensing imagery used for fDOM model development in this study are available through the European Space Agency Copernicus Data Space Ecosystem Sentinel-2 Repository (Copernicus Sentinel 2 2021; https://doi.org/10.5270/S2_-742ikth). Imagery data are downloaded using the API for batch processing. The ACOLITE (v20180327.0) software (Vanhellemont, 2021) used to perform Sentinel-2 atmospheric correction is freely available via the ACOLITE GitHub (https://github.com/acolite?tab=overview&from=2020-12-01&to=2020-12-31). ACOLITE v20180327.0 was the most recent stable version of ACOLITE available at the time the Sentinel-2 images were processed. More recent versions are now available, along with clear guidance on installing and using ACOLITE. Measurements of fDOM collected at USGS continuous monitoring stations, and DOC measurements from the ship-based surveys are available via the USGS Water Data for the Nation (U.S. Geological Survey, 2021) (https://waterdata.usgs.gov/nwis). Measurements of fDOM, attenuation, and absorption collected during ship-based mapping cruises are

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available in a USGS ScienceBase data release (Fleck et al., 2023). The historic measurements of attenuation and absorption collected in Lake Erie and the Sacramento-San Joaquin Delta are available through the SeaBASS repository (Werdell et al., 2003) using a geographical search for absorption (a) and attenuation (c) data measured in California between 1 January 2014 and 1 October 2015. There are two files of data from the National Research Laboratory (NRL/RIO-SFE/SFE-1_2014 05/archive/SFE_1_IOP_Ackleson_122216.sb and NRL/RIO-SFE/SFE-3_2015-03/archive/SFE_3_IOP_Ackleson_122216.sb) available from SeaBASS (https://seabass.gsfc.nasa.gov/search#bio).

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